

Failure Modes and Constraint Coverage in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

Cara Tonucci

Independent Researcher

April 24, 2026 — Version 1

Cara.Tonucci@gmail.com

ORCID: <https://orcid.org/0009-0003-9048-5821>

Abstract

Failure modes and constraint coverage in Dimensional Human Field Theory (DHFT) define the conditions under which system integrity may be violated and the corresponding constraint mechanisms that prevent such violations.

Each failure mode is mapped to a distinct constraint, establishing complete coverage across system definition, access, use, classification, and exposure.

Failure mode mapping governs system integrity and does not define system mechanics.

I. Scope

Failure mode analysis governs:

- conditions that compromise system integrity
- constraint coverage across governance artifacts
- preservation of structural admissibility across representations

It does not define system mechanics.

II. Failure Mode Definition

A failure mode is a condition under which structural integrity, classification, or admissible use of DHFT is violated.

Failure modes are independent and non-interchangeable.

III. Failure Modes

Non-Admissible Structure

Violation of invariant variables or structural relations.

Incorrect Entry

Violation of dependency ordering or admissibility conditions during system access.

Misuse

Application outside structural constraints or use in interpretive, diagnostic, or narrative contexts.

Misclassification

Classification based on application context, domain similarity, or interpretive framing.

Interface Collapse

Inference of system definition from outputs, applications, or points of exposure rather than invariant structure.

Authority Drift

Stabilization of altered definitions, relations, or classifications as canonical through repeated use or attribution.

IV. Constraint Coverage Mapping

Each failure mode is prevented by a corresponding constraint mechanism:

- Non-Admissible Structure → Admissibility Conditions
- Incorrect Entry → Entry Protocol
- Misuse → Boundary of Use
- Misclassification → Classification Constraints
- Interface Collapse → Interface Protocol and Classification Constraints

V. Independence of Constraints

Constraint mechanisms are:

- independent
- non-redundant
- non-substitutable

Removal or modification of any constraint introduces an unprotected failure mode.

VI. Coverage Completeness

The set of constraint mechanisms provides complete coverage across:

- structural admissibility
- system access
- system use
- system classification
- system exposure

The defined set of failure modes is exhaustive at the system level.

All violations reduce to one or more defined failure modes.

No additional constraint or failure categories exist.

VII. Non-Interpretive Condition

Failure modes and constraints are defined structurally and do not:

- assign intent
- infer motive
- produce interpretation

They define admissibility conditions only.

VIII. Canonical Statement

DHFT integrity is preserved through a set of independent constraint mechanisms, each mapped to a distinct failure mode.

System admissibility requires that all constraint conditions remain satisfied.

IX. System Reference

The system referenced in this document is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical notes.

X. Copyright

This document is an original work of authorship.

No part of this publication shall be reproduced, distributed, or transmitted in any form or by any means without prior written permission, except for brief quotations for academic or scholarly use.

This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

© 2026 Cara Tonucci. All rights reserved.